

Q1

$$a) \left(\overset{5}{1} \overset{4}{0} \overset{3}{0} \overset{2}{1} \overset{1}{1} \overset{0}{1} . \overset{-1}{1} \overset{-2}{1} \overset{-3}{0} \overset{-4}{1} \overset{-5}{1} \right)_2$$

$$= 1 \times 2^5 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-4} + 1 \times 2^{-5}$$

$$= (39.84375)_{10}$$

$$c) \left(\overset{2}{7} \overset{1}{9} \overset{0}{B} . \overset{-1}{2} \overset{-2}{F} \overset{-3}{G} \right)_{17}$$

$$= 7 \times 17^2 + 9 \times 17^1 + 11 \times 17^0 + 2 \times 17^{-1} + 15 \times 17^{-2} + 16 \times 17^{-3}$$

$$= (2187.172807)_{10}$$

Q2

a) $(107.0425)_{10}$

$2 \overline{) 107}$	
$2 \overline{) 53} \rightarrow 1$	
$2 \overline{) 26} \rightarrow 1$	
$2 \overline{) 13} \rightarrow 0$	
$2 \overline{) 6} \rightarrow 1$	
$2 \overline{) 3} \rightarrow 0$	
$1 \overline{) 1} \rightarrow 1$	
$0 \rightarrow 1$	

LSB

↑

MSB

MSB

	0.0425
	x 2
0	.085
	x 2
0	.17
	x 2

$$\begin{array}{c|c} \downarrow \text{LSB} & 0.34 \end{array}$$

So,

$$(107.0425)_{10} = (1101011.000\dots)_2$$

$$b) (6045)_7 \rightarrow (?)_{10}$$

$$\begin{array}{cccc} 3 & 2 & 1 & 0 \\ 6 & 0 & 4 & 5 \end{array}$$

$$= 6 \times 7^3 + 0 \times 7^2 + 4 \times 7^1 + 5 \times 7^0$$

$$= (2091)_{10}$$

$$\begin{array}{r} 15 \overline{) 2091} \\ 15 \overline{) 139} \rightarrow 6 \\ 15 \overline{) 9} \rightarrow 4 \\ 0 \rightarrow 9 \end{array}$$

$\uparrow$ LSB
 $\downarrow$ MSB

So,

$$(6049)_7 = (946)_{15}$$

$$\begin{array}{r} Q_4 \\ \hline \end{array}$$

a)

$$\begin{array}{r} 1 \\ 4 \text{ E B } 4 \\ + 9 \text{ B} \\ \hline \end{array}$$

$$B + 9 = 20$$

$$20 \div 17$$

$$= 3$$

Addition $(4 \text{ F } 3 \text{ F})_{17}$

Subtraction:

$$4 \text{ E B } 4$$

$$\begin{array}{r} - 9 \text{ B} \\ + 1 \end{array}$$

$$\begin{array}{r} (4 \text{ E } 1 \text{ A})_{17} \end{array}$$

$$4 + 17 = 21$$

$$21 - B = A$$

Multiplication:

+2 +7 +5 +2
+9 +7 +2

+3 4 E B 4

x 9 B

1
3 2 8 4 A

+ 9 C 6 2 x

(2 C F 7 6 A)₁₇

Red marked carries are from second digit multiplication

$$9 \times 4 = 36 \text{ / } 17 = 2$$

$$11 \times 9 = 99 + 2 = 101 \text{ / } 17 = 6$$

$$9 \times 14 = 126 + 5 = 131 \text{ / } 17 = C$$

$$(B) \quad 11 \times 4 = 44$$

$$44 \text{ / } 17$$

$$= A$$

$$11 \times 11$$

$$= 121 + 2$$

$$= 123$$

$$123 \text{ / } 17$$

$$= 4$$

$$11 \times 14$$

$$= 154 + 7$$

$$= 161 \text{ / } 17$$

$$= 8$$

$$11 \times 4$$

$$= 44 + 9$$

$$= 53 \text{ / } 17$$

$$= 2$$

Q5

$$(4432)_5 / (13)_5$$

$$13 \overline{) 4432(0302}$$

$$\begin{array}{r} 0 \\ \hline 44 \\ - 44 \\ \hline 3 \\ 0 \\ \hline 32 \\ 31 \\ \hline 1 \end{array}$$

Number Table

$$\begin{array}{r} 13 \\ \times 1 \\ \hline 13 \end{array}$$

+1

$$\begin{array}{r} 13 \\ \times 2 \\ \hline 31 \end{array}$$

+1

$$\begin{array}{r} 13 \\ \times 3 \\ \hline 44 \end{array}$$

So,

quotient $\rightarrow (302)_5$

remainder $\rightarrow (1)_5$

Q6

$(F1)_{16}$

$$\begin{aligned} \hookrightarrow 15 \times 16^1 + 1 \times 16^0 \\ = (241)_{10} \end{aligned}$$

Price for two RAM,

$$(241 \times 2) = (482)_{10}$$

GPU price,

$$\begin{aligned} & \overset{10}{1} \overset{9}{0} \overset{8}{1} \overset{7}{1} \overset{6}{1} \overset{5}{1} \overset{4}{0} \overset{3}{0} \overset{2}{1} \overset{1}{1} \overset{0}{1} \\ & (10111100111)_2 \\ & = 1 \times 2^{10} + 1 \times 2^8 + 1 \times 2^7 + 1 \times 2^6 \end{aligned}$$

$$+ 1 \times 2^5 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$= (1511)_{10}$$

Total money spent,

$$(1511 + 482) = (1993)_{10}$$

Friend has given,

$$(3376)_8$$

$$= 3 \times 8^3 + 3 \times 8^2 + 7 \times 8^1 + 6 \times 8^0$$

$$= (1790)_{10}$$

You will be short,

$$(1993 - 1790) = (203)_{10}$$

dollars

Q7

a) $(472.2097)_{10}$

Excess - 6 : $9 + 6 = 15$

$\Downarrow$
 1111

4-bit code

So,

Let's add 6 with each digit (You can do it in decimal and then convert to binary to get corresponding codes or you can do it after finding the BCD for each digit and then add $6 (110)_2$ with each code)

$$4 + 6 = 10 \rightarrow 1010$$

$$7 + 6 = 13 \rightarrow 1101$$

$$2 + 6 = 8 \rightarrow 1000$$

$$2 + 6 = 8 \rightarrow 1000$$

$$0 + 6 = 6 \rightarrow 0110$$

$$9 + 6 = 15 \rightarrow 1111$$

$$7 + 6 = 13 \rightarrow 1101$$

So,

$$(101011011000.1000011011111101)_{\text{Excess-6}}$$

b)

$$(1010011011011001010.01001001)_{\text{Excess-3}}$$

Excess-3: $9 + 3 = 12 \rightarrow 1100$
4-bit
code

So let's group 4-bit together from LSB and then subtract 3 (11_2) from each to find BCD (can be done by converting to decimal first and then subtract 3 in decimal as well (just like question a))

(01010011011011001010.01001001)

$$\begin{array}{r} 0101 \\ - 11 \\ \hline 0010 \end{array}$$

$$\begin{array}{r} 0011 \\ - 11 \\ \hline 0000 \end{array}$$

$$\begin{array}{r} 0110 \\ - 11 \\ \hline 0011 \end{array}$$

$$\begin{array}{r} 1010 \\ - 11 \\ \hline 0111 \end{array}$$

$$\begin{array}{r} 1100 \\ - 11 \\ \hline 1001 \end{array}$$

$$\begin{array}{r} 0100 \\ - 11 \\ \hline 0001 \end{array}$$

$$\begin{array}{r} 1001 \\ - 11 \\ \hline 0110 \end{array}$$

So the BCD is,

$(\underbrace{0010}_{2}\underbrace{0000}_{0}\underbrace{0011}_{3}\underbrace{1001}_{9}\underbrace{0111}_{7}.\underbrace{0001}_{0.16})_{BCD}$

$$= (20397.16)_{10} \rightarrow (?)_8$$

8	20397	
8	2549 → 5	
8	318 → 5	
8	39 → 6	
8	4 → 7	
	0 → 4	

LSB

MSB

MSB

0.16
 $\times 8$

1	. 28 × 8
2	. 24 × 8

$$\begin{array}{c} \downarrow \\ \text{LSB} \end{array} \quad \begin{array}{c|c} 1 & .92 \end{array}$$

So the converted number,

$$(47655.121\dots)_8$$

$$\underline{\text{Q8}}$$

a) $(+147)_{10}$ in 9-bit

$$(147)_{10} \rightarrow (10010011)_2$$

$(147)_{10}$ in 9-bit,

$$(\boxed{0}^{ \text{sign bit} } 10010011)_2$$

This is $(+147)_{10}$ in

9-bit sign and magnitude

format and also in

1's and 2's complement
as complement cannot be done
for positive numbers

b) $(-193)_{10}$

$(193)_{10} \rightarrow 110000001$

In 9-bit

$(011000001)_2$

In sign and magnitude,

$(111000001)_2$

sign

1's complement,

011000001

$(100111110)_2$ (1's)

2's complement

$$\begin{array}{r} 100111110 \\ +1 \\ \hline (100111111)_2 \end{array}$$

Q9

$(-25)_{10} + (-97)$ in 7-bit

2's complement

$(25)_{10}$

$\rightarrow (11001)_2$

In 7-bit,

0011001

1100110

+1

1100111

(1's)

(2's)

$(97)_{10}$

↳ $\begin{matrix} 64 & 32 & 16 & 8 & 4 & 2 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 \end{matrix}$ (7-bit magnitude)

1's,

00 1 1 1 1 0

+ 1

00 1 1 1 1 1 (2's)

Note: There is mismatch in sign as the magnitude of $(97)_{10}$ takes 7-bit, there is no place to keep the sign

Let's perform the addition normally,

$$\begin{array}{r}
 \overset{1}{1} \overset{1}{1} \overset{1}{1} \overset{1}{1} \overset{1}{1} \overset{1}{1} \overset{1}{1} \\
 \underline{1100111} \\
 + 0011111 \\
 \hline
 \boxed{1}0000110
 \end{array}$$

discard

Result $\rightarrow (0000110)_2$

Expected Result $\rightarrow (-25) + (-97)$

$$= -122$$

$$= -ve$$

calculated Result $\rightarrow +ve$

So,

There is overflow